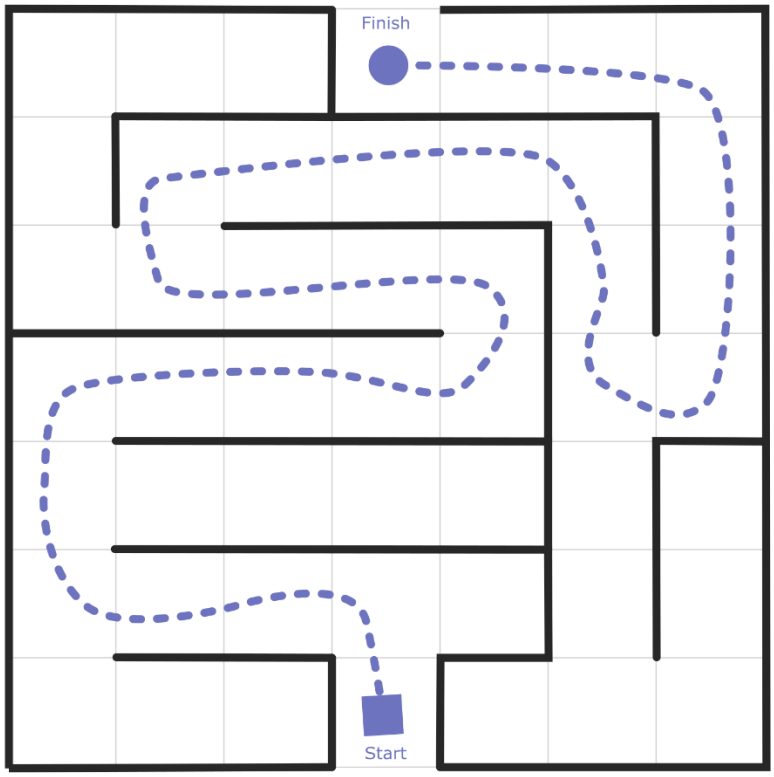
وزارة الدفاع
MINISTRY OF DEFENCE

DRONE CHALLENGE

Challenge 1 – Navigate the Maze

Premises



- *n*-by-*n* maze
- Pre-defined end square: The target square is known.
- Pre-defined start square
- Drone is placed into the start square at a pre-defined bearing (north: camera facing in the positive y-direction)
- The maze configuration (i.e. walls inside the maze) is unknown to the competitors

Objective

Find the path between start and finish.

The drone must use its proximity sensors to detect maze barriers.

The drone must use QR code carpet for navigation.

Challenge Execution

Before Execution

- All teams are required to submit their code the **evening before the first day of UMEX** (19th of January).
    - This code "freeze" will allow the challenge to test the code without knowing the inner configuration of the maze. **We will not allow any changes to the algorithm during UMEX.**
    - Teams will be allocated a dedicated laptop and drone during the workshop. They will save their final competition code on that laptop.
- All teams must use the python and drone SDK version installed on their allocated laptop. They may add additional python libraries (pip install).
- All teams will be allocated a fixed timeslot to execution the challenge on the day of UMEX. Teams are required to arrive 30 minutes **before** their timeslot to meet their judge and check the laptop and drone hardware.

During Execution

The execution is split into two phases

1. Discovery Phase
2. Scoring Phase



Discovery Phase

- Each team will be given 10 minutes to **discover the maze**. During this time, the drone can autonomously fly through the maze. **The code can remember the maze setup** (barrier locations).

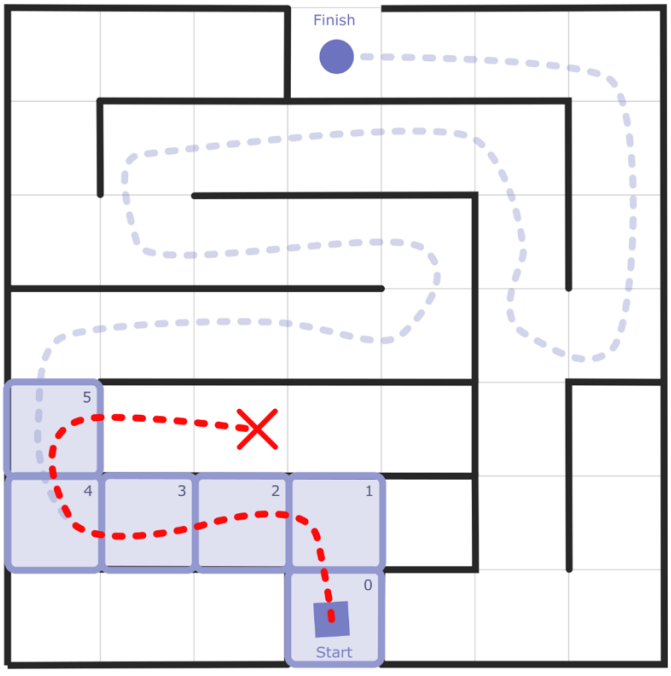
- There is no scoring in this phase.
- Teams can restart flying from any point inside the maze at any time.

Racing Phase

- After *Discovery Phase* has concluded, teams have 10 minutes **to solve the maze**: To reach the target square from the start square.
- The **timer will start** when the drone leaves the ground.
- The **timer will stop** when the drone touches the ground in the target square.
- Within the 10 minutes, teams can **restart** from the start square **as often as they like**.
- The best score of all flight executions will be taken flight executions.
- Every time the drone lands or crashes outside the target square, the drone restart in the start square and the timer will reset.

Scoring

1. Time taken from *Start* to *Finish* (primary score)
2. Points for passing every square of the correct path (furthest progression – secondary score)



- The team captain will be required **to sign-off** the result after challenge execution
- Any dispute (refusal to sign-off) will be escalated to the Drone Challenge committee for fair mediation.